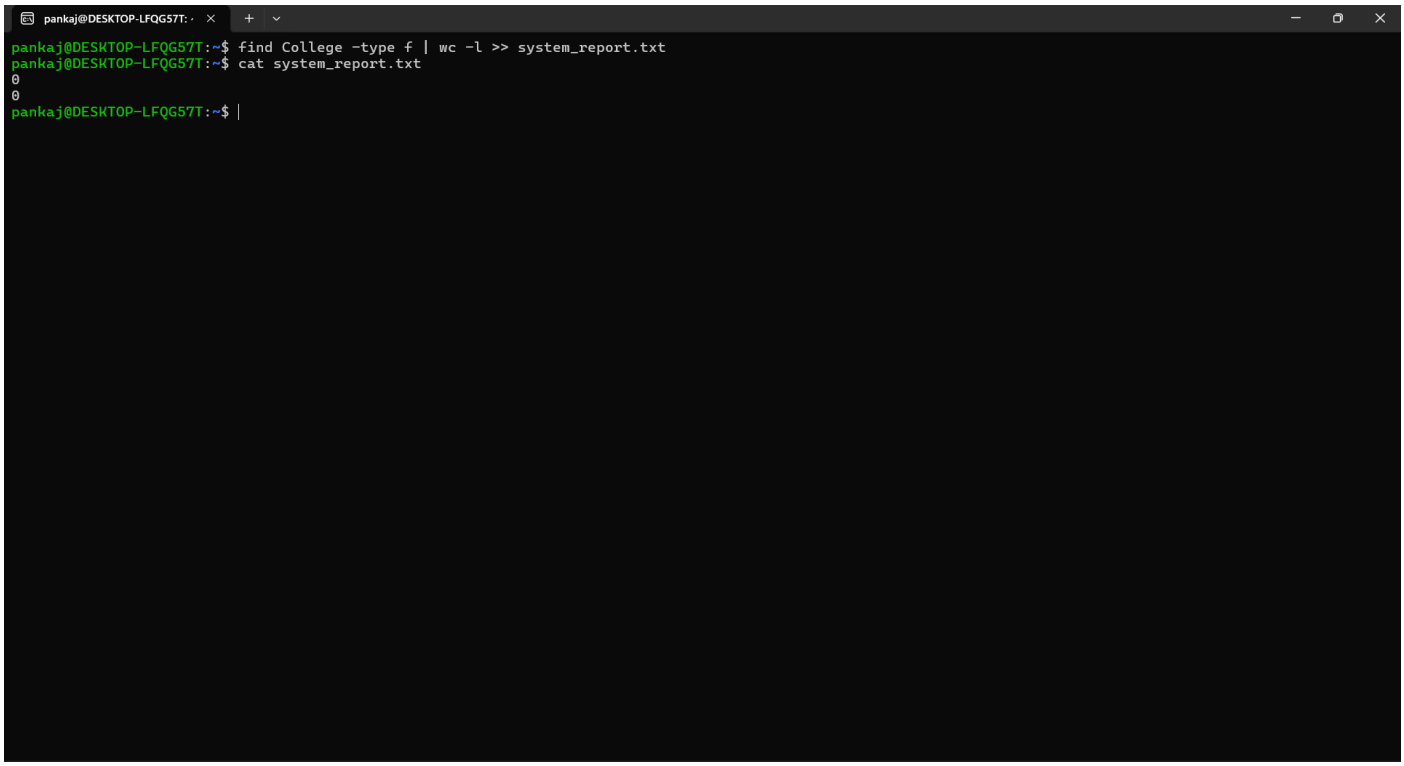
Question 7

Generate a system/directory report using find, du, df, stat, wc, sort, head, tail, uname, hostname, history, free, ps, top with screenshots and summary.

Step 1 – Count files in the College hierarchy

Commands used: find College -type f | wc -l >> system_report.txt.

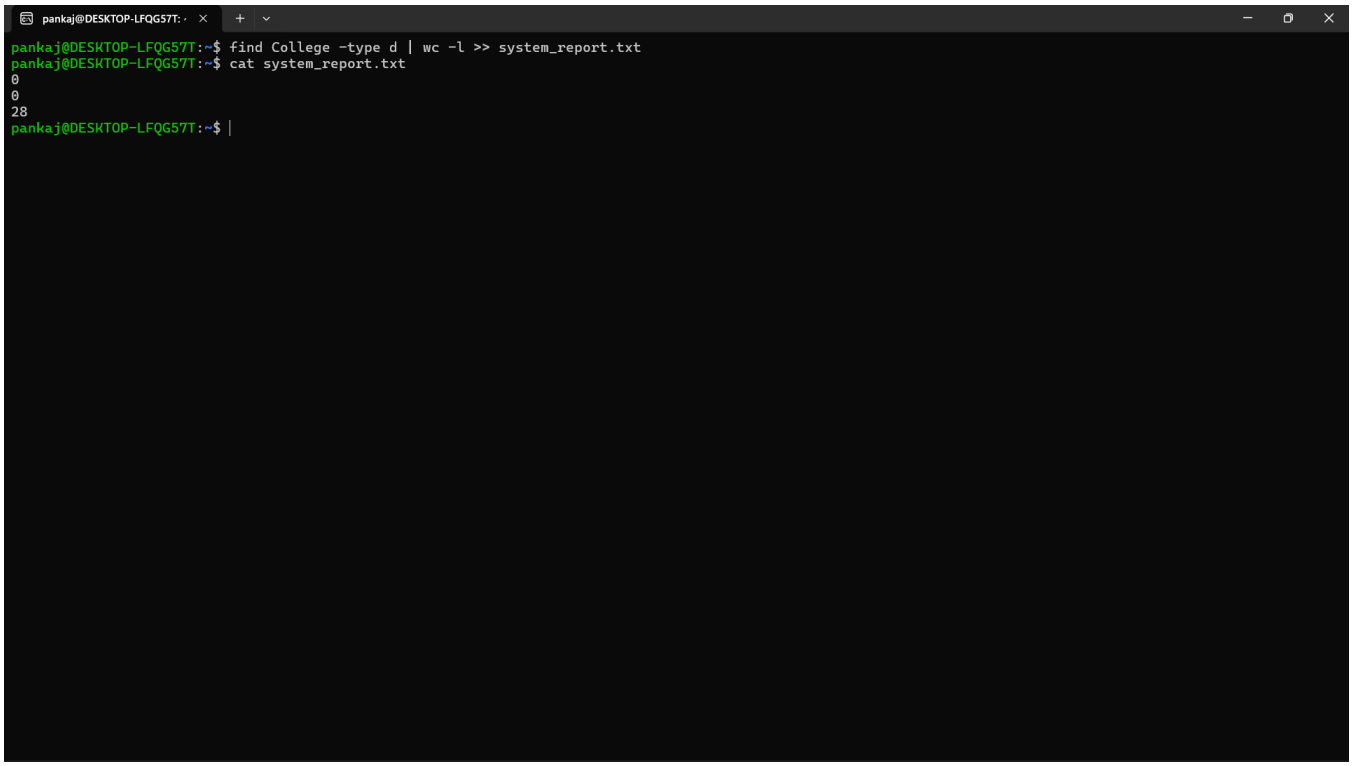
We counted all files inside the College hierarchy and appended the resulting count to system_report.txt.



Step 2 – Count directories in the College hierarchy

Commands used: find College -type d | wc -l >> system_report.txt.

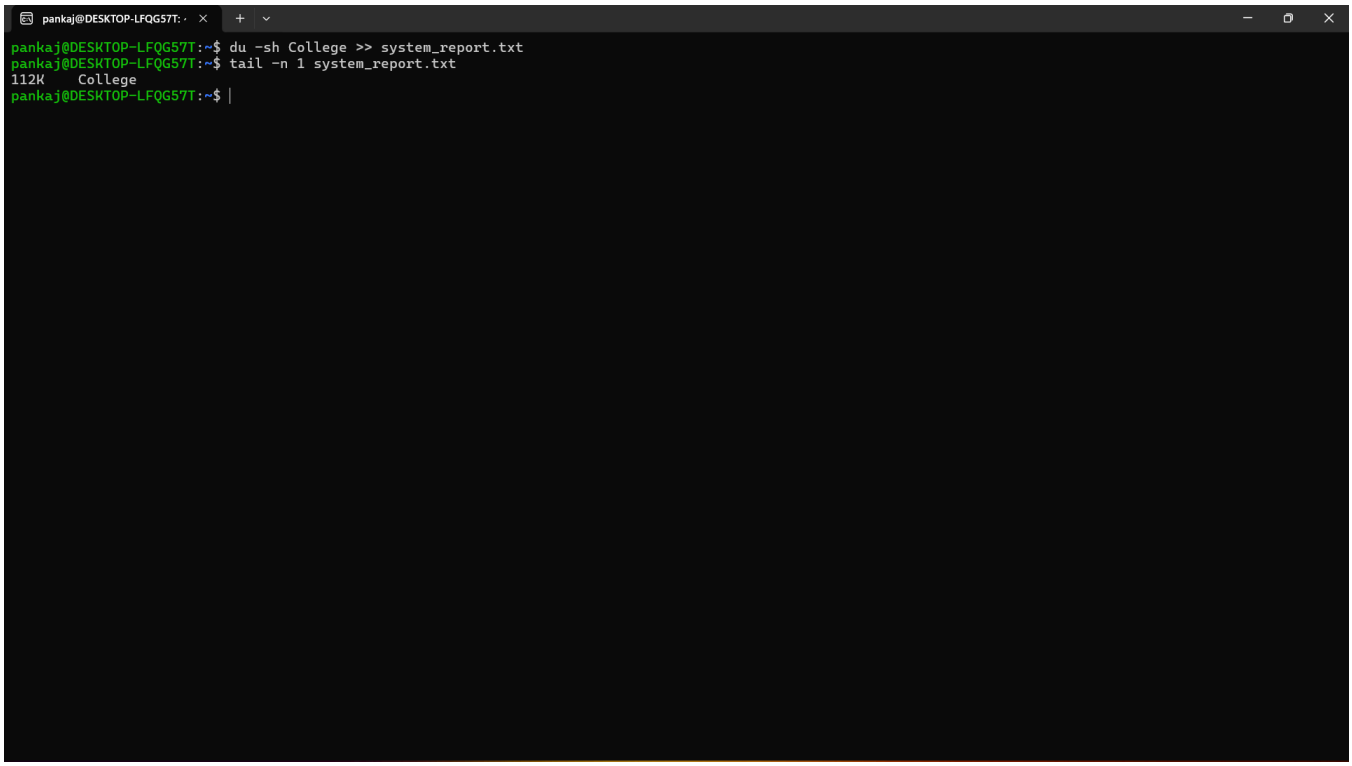
We counted all directories inside the College hierarchy and appended the resulting count to system_report.txt.



Step 3 – Check College directory disk usage

Commands used: du -sh College >> system_report.txt, tail -n 1 system_report.txt.

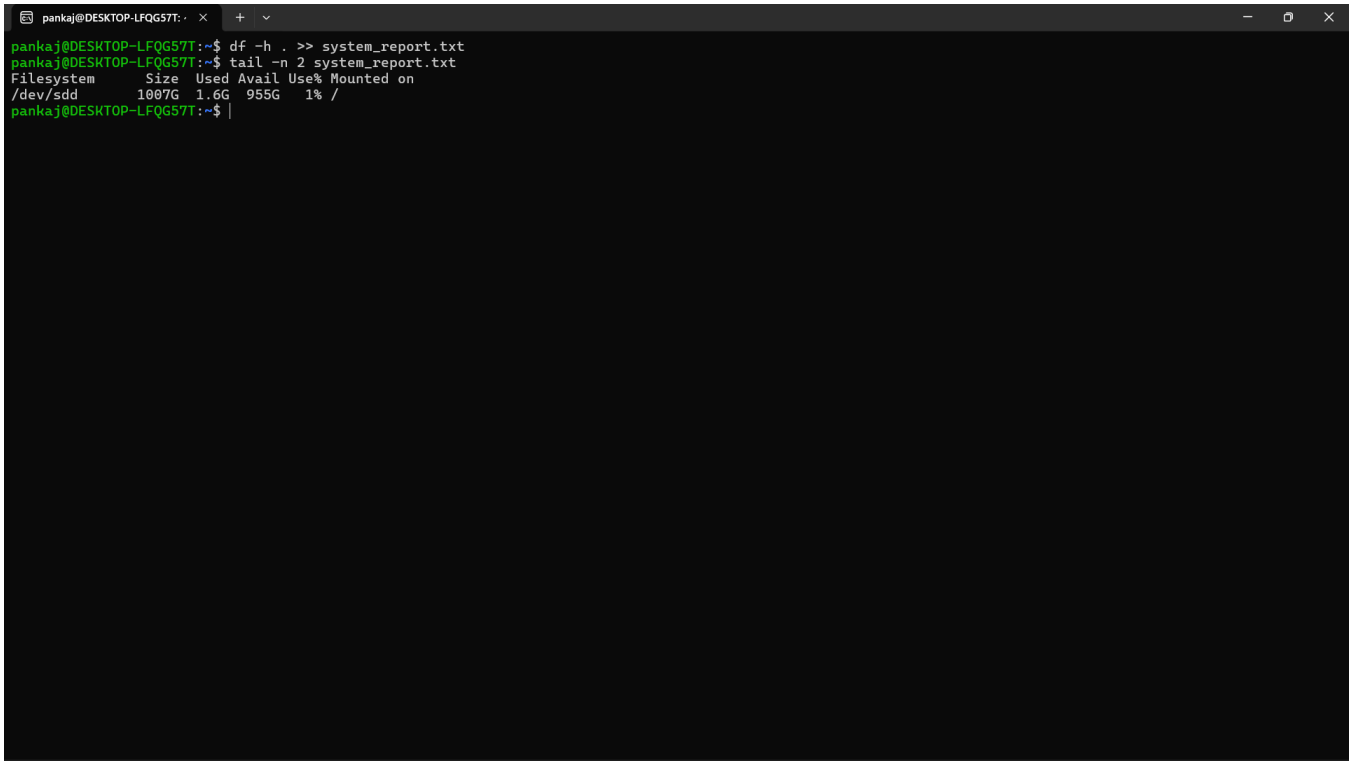
We calculated the total human-readable disk usage of the College directory and appended the result to the report.



Step 4 – Check filesystem capacity

Commands used: df -h . >> system_report.txt, tail -n 2 system_report.txt.

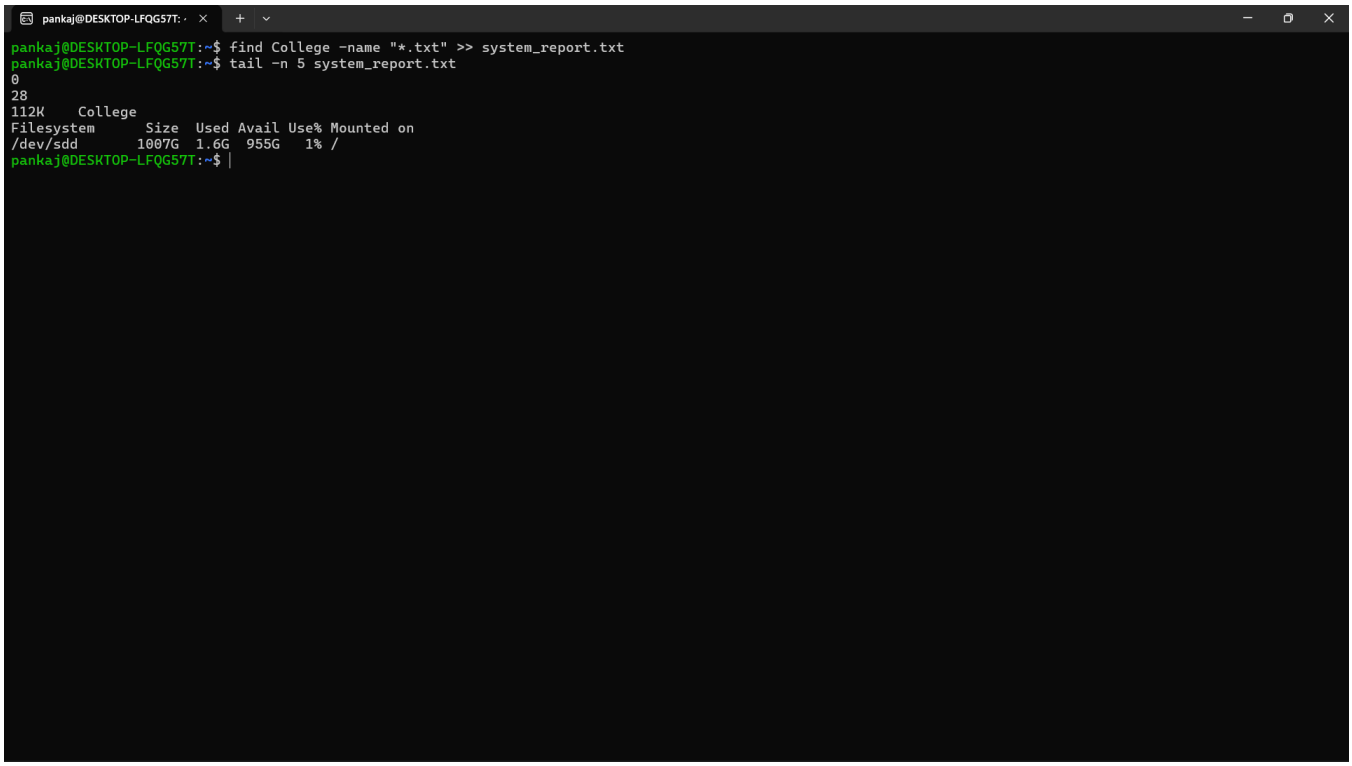
We displayed the filesystem capacity for the current mount in human-readable form and appended the result to the report.



Step 5 – Find text files

Commands used: find College -name "*.txt" >> system_report.txt, tail -n 5 system_report.txt.

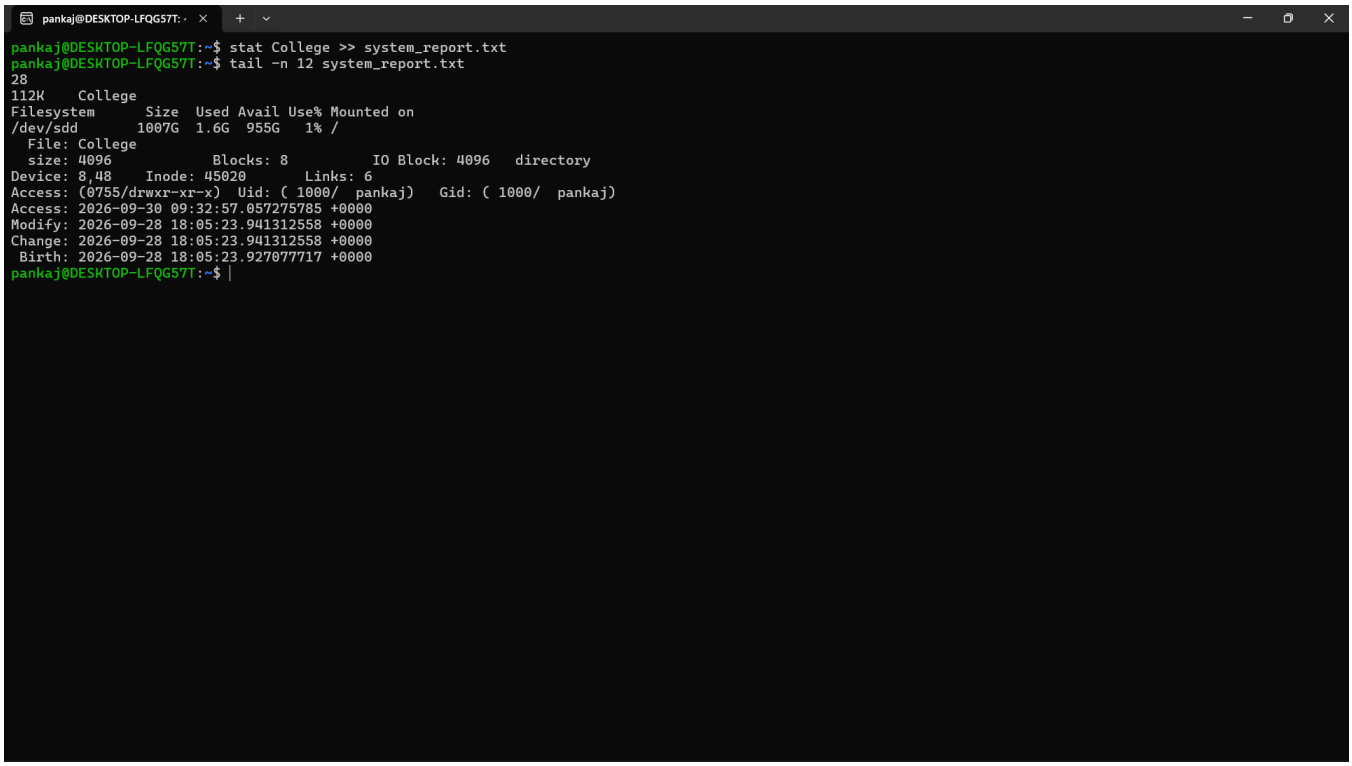
We searched the College hierarchy for .txt files and appended the matching paths to the report.



Step 6 – Display College metadata

Commands used: stat College >> system_report.txt, tail -n 12 system_report.txt.

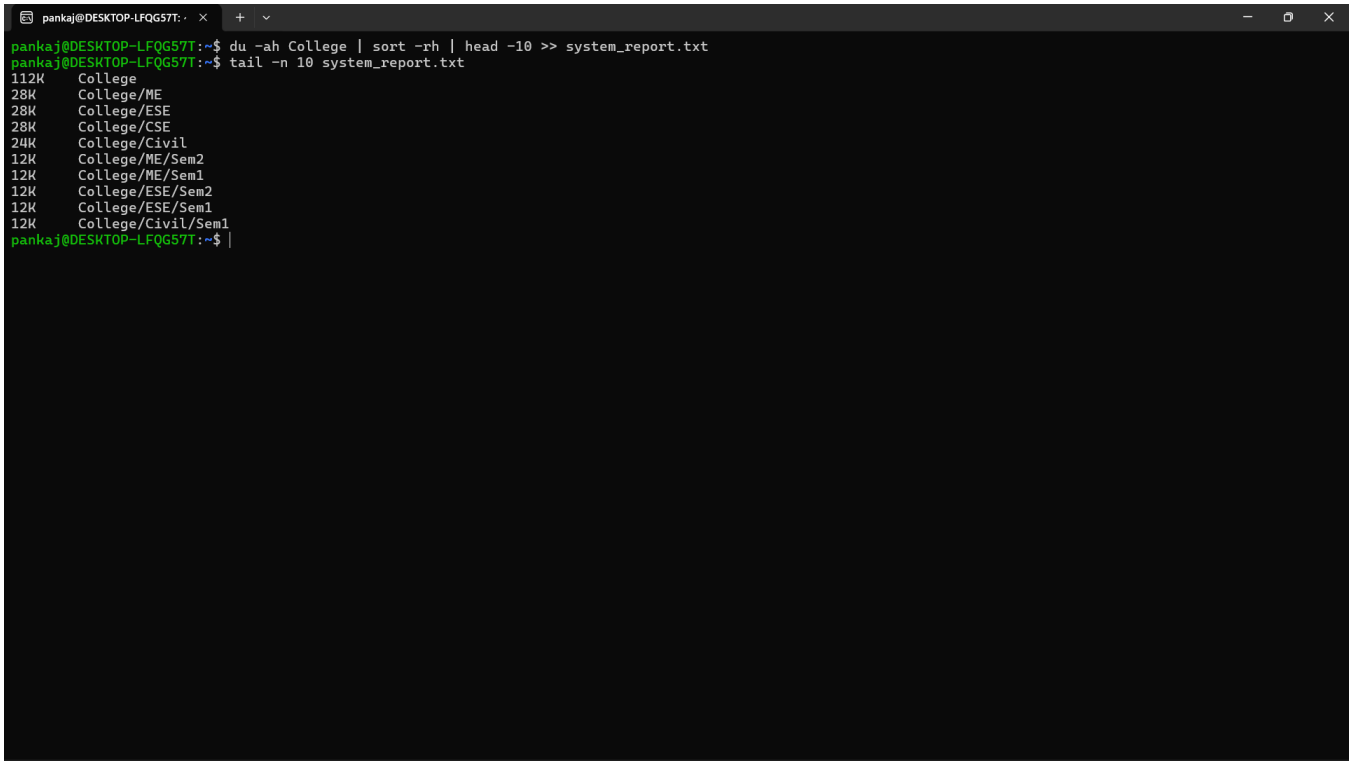
We collected metadata for the top-level College directory, including permissions, timestamps, size and inode information.



Step 7 – Rank the largest College entries

Commands used: du -ah College | sort -rh | head -10 >> system_report.txt.

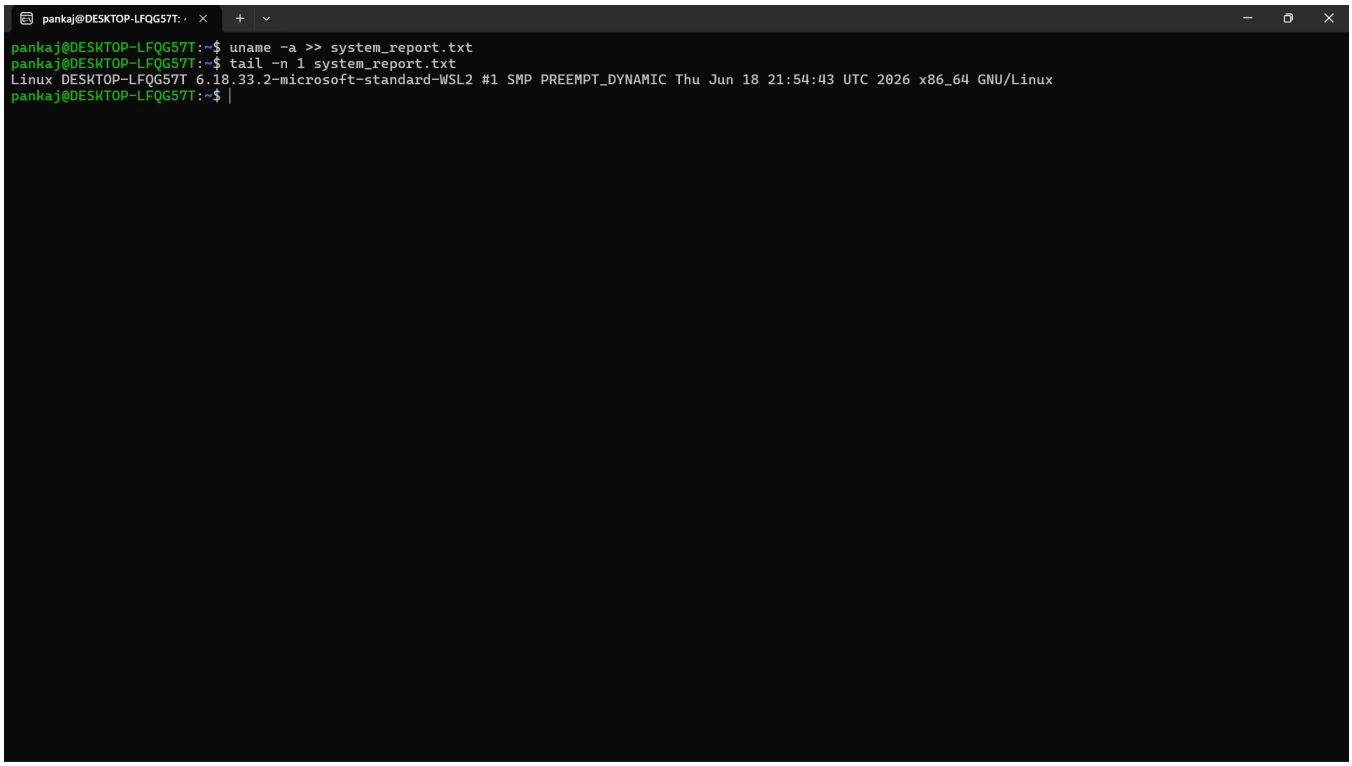
We calculated sizes for entries in College, sorted them in reverse human-readable order and recorded the ten largest entries.



Step 8 – Collect system information

Commands used: uname -a >> system_report.txt.

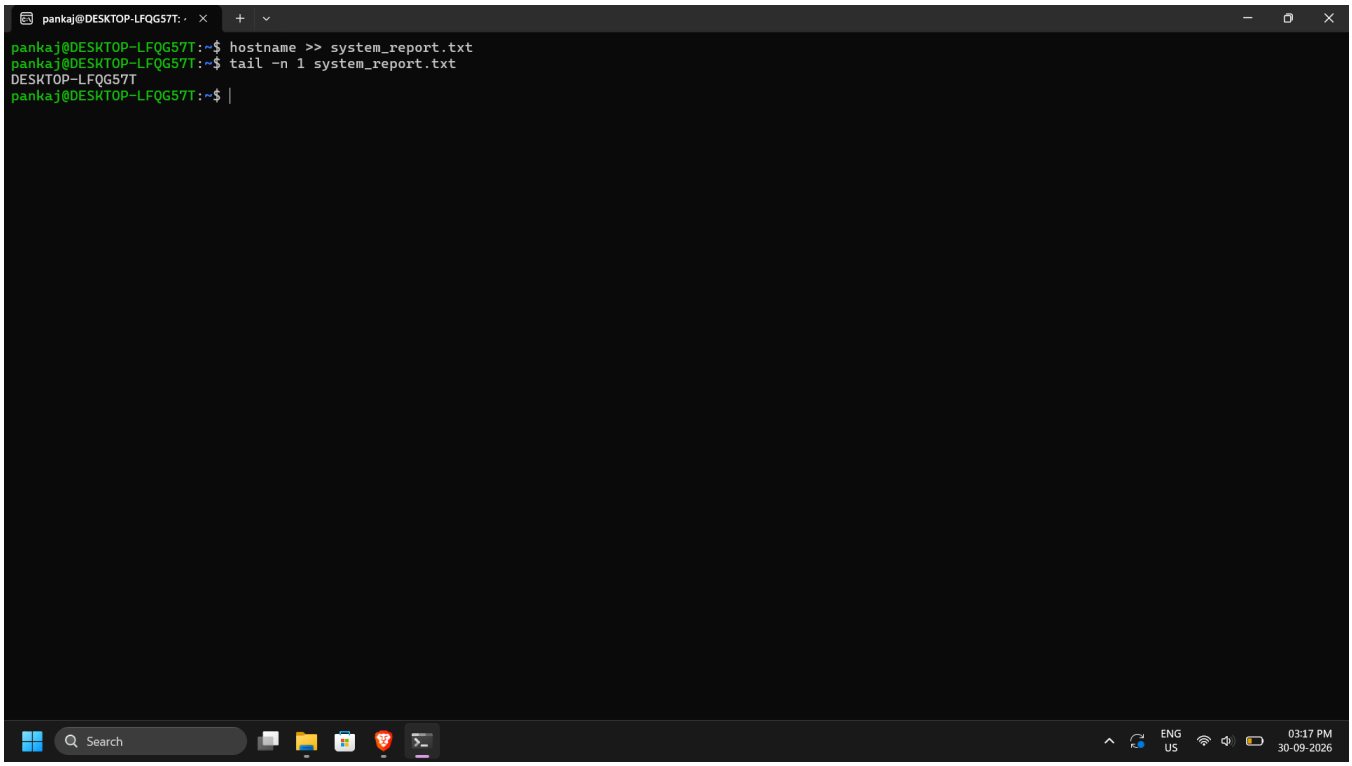
We recorded kernel, operating-system and architecture information in the report.



Step 9 – Display the hostname

Commands used: hostname >> system_report.txt.

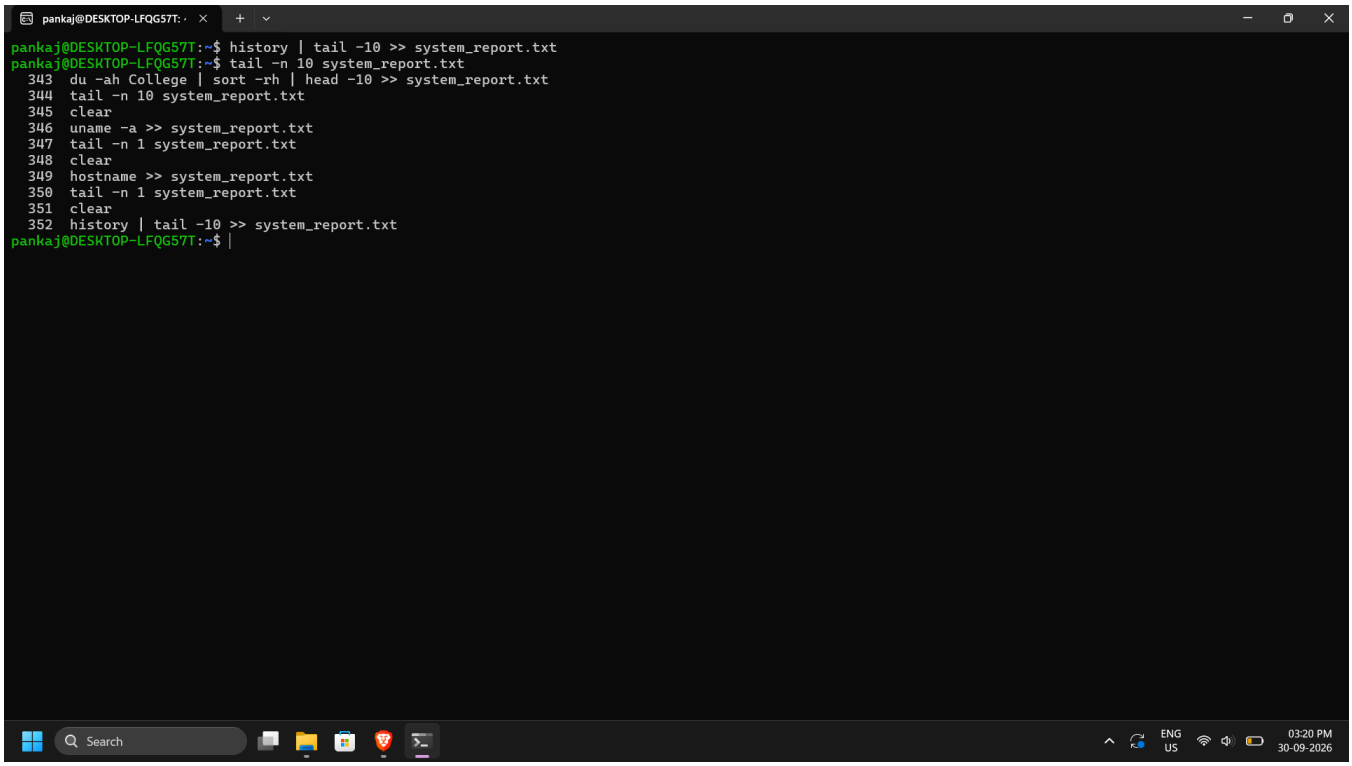
We recorded the machine hostname in the consolidated report.



Step 10 – Record recent shell history

Commands used: history | tail -10 >> system_report.txt.

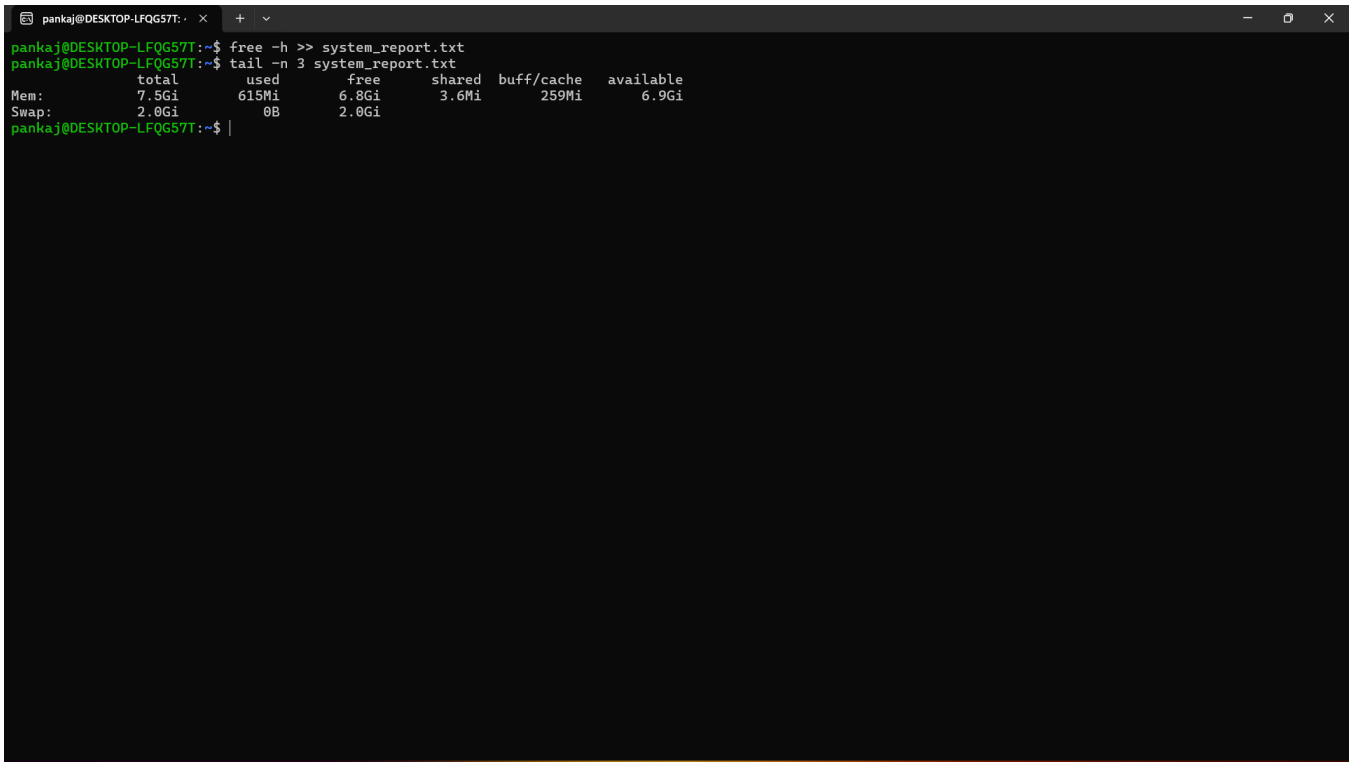
We appended the ten most recent shell commands from the current session to the report.



Step 11 – Check memory and swap usage

Commands used: free -h >> system_report.txt.

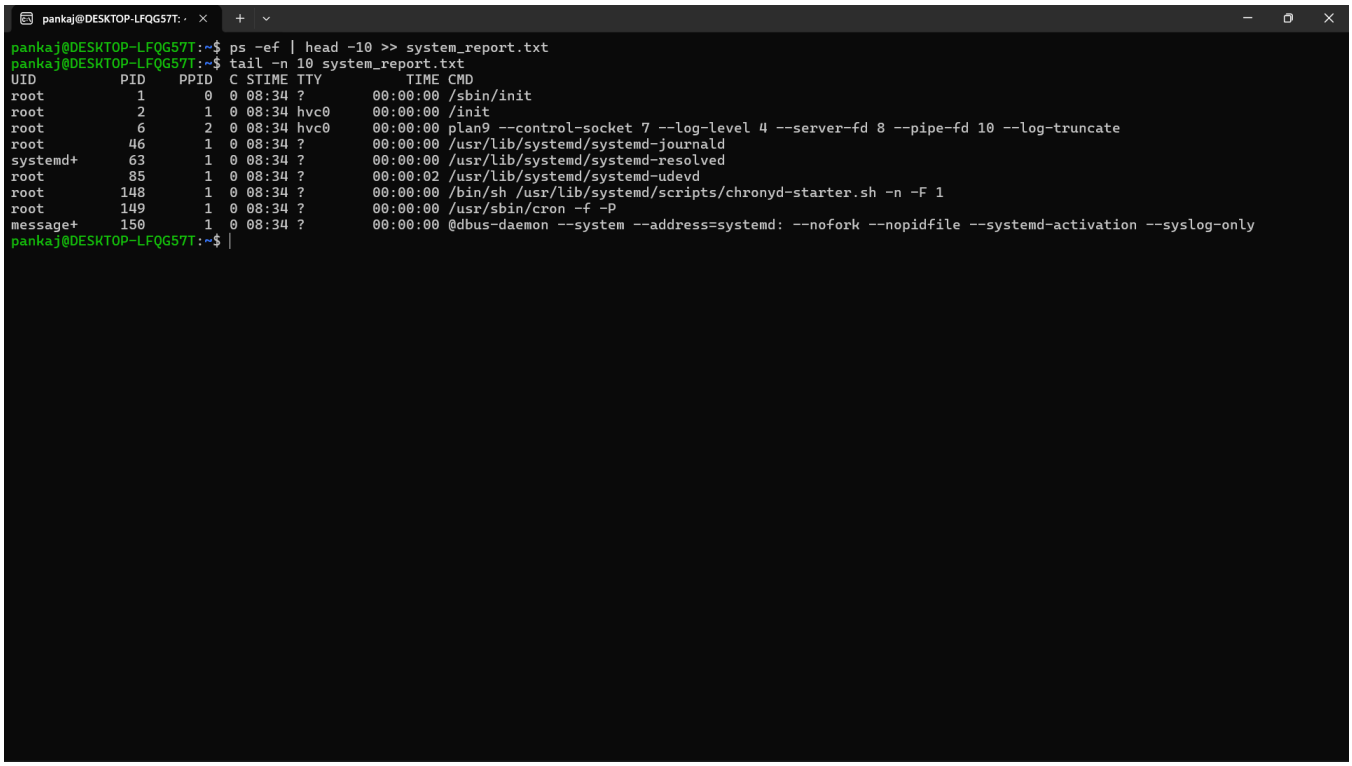
We recorded current memory and swap usage in human-readable form.



Step 12 – Display running processes

Commands used: ps -ef | head -10 >> system_report.txt.

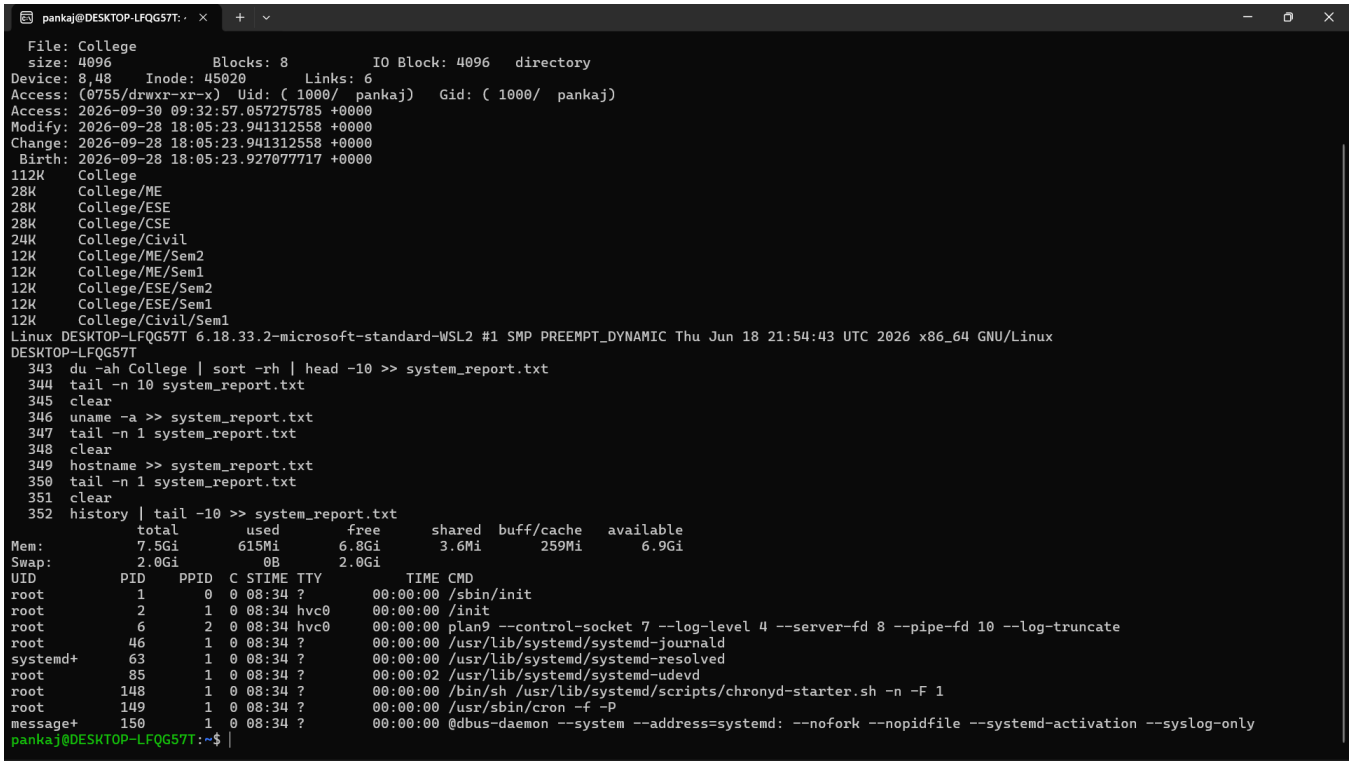
We recorded a snapshot of the first ten running processes in the report.



Step 13 – Display the completed system report

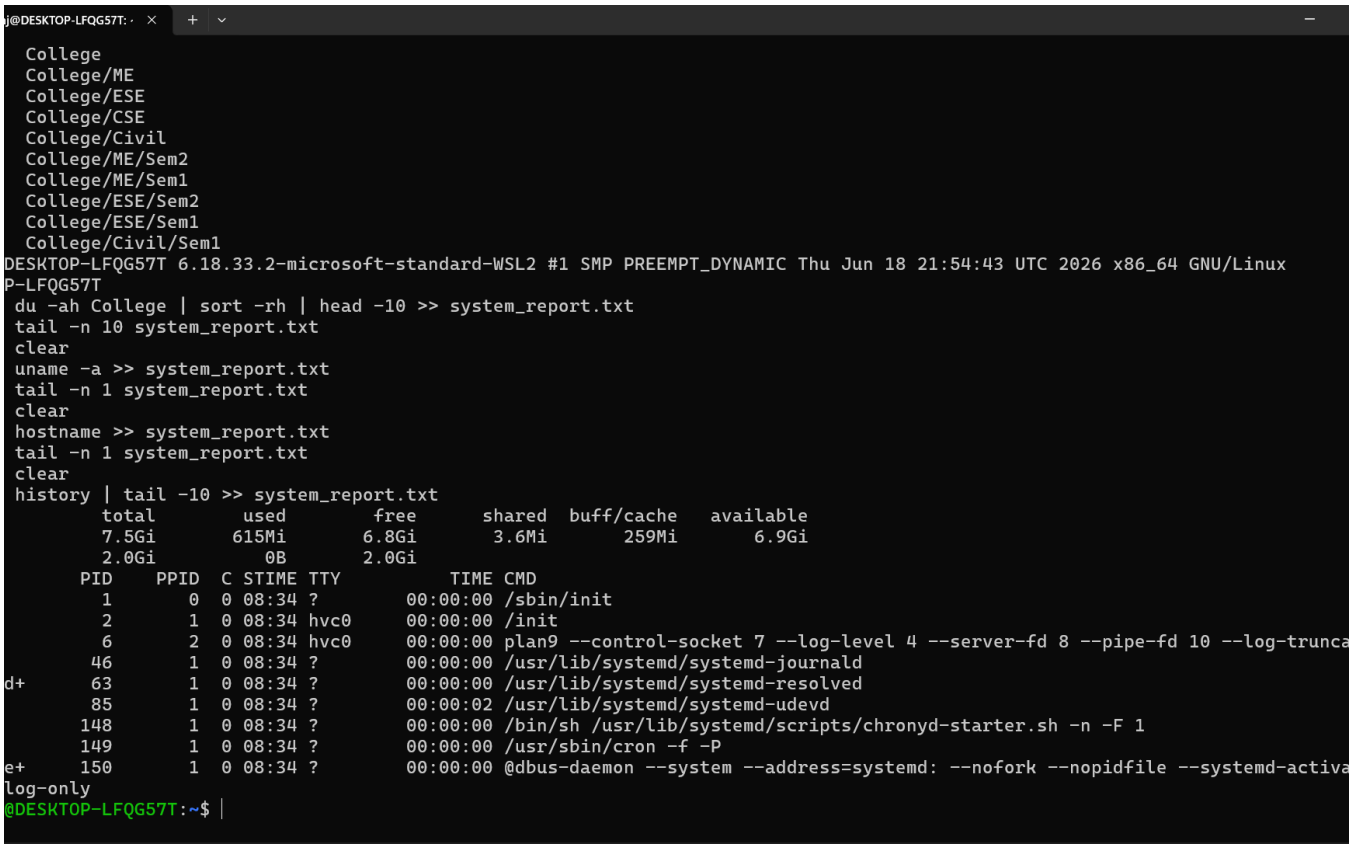
Commands used: cat system_report.txt.

We displayed the completed consolidated report. The output was longer than one terminal screen, so the final evidence is shown across two screenshots.



Step 14 – Display the completed system report (continued)

Continuation of the final system_report.txt output.


j@DESKTOP-LFQG57T:
College
College/ME
College/ESE
College/CSE
College/Civil
College/ME/Sem2
College/ME/Sem1
College/ESE/Sem2
College/ESE/Sem1
College/Civil/Sem1
DESKTOP-LFQG57T 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
P-LFQG57T
du -ah College | sort -rh | head -10 >> system_report.txt
tail -n 10 system_report.txt
clear
uname -a >> system_report.txt
tail -n 1 system_report.txt
clear
hostname >> system_report.txt
tail -n 1 system_report.txt
clear
history | tail -10 >> system_report.txt
total used free shared buff/cache available
7.5Gi 615Mi 6.8Gi 3.6Mi 259Mi 6.9Gi
2.0Gi 0B 2.0Gi
PID PPID C STIME TTY TIME CMD
1 0 0 08:34 ? 00:00:00 /sbin/init
2 1 0 08:34 hvc0 00:00:00 /init
6 2 0 08:34 hvc0 00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-trunca
46 1 0 08:34 ? 00:00:00 /usr/lib/systemd/systemd-journald
d+ 63 1 0 08:34 ? 00:00:00 /usr/lib/systemd/systemd-resolved
85 1 0 08:34 ? 00:00:02 /usr/lib/systemd/systemd-udevd
148 1 0 08:34 ? 00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
149 1 0 08:34 ? 00:00:00 /usr/sbin/cron -f -P
e+ 150 1 0 08:34 ? 00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activa
log-only
@DESKTOP-LFQG57T:~$